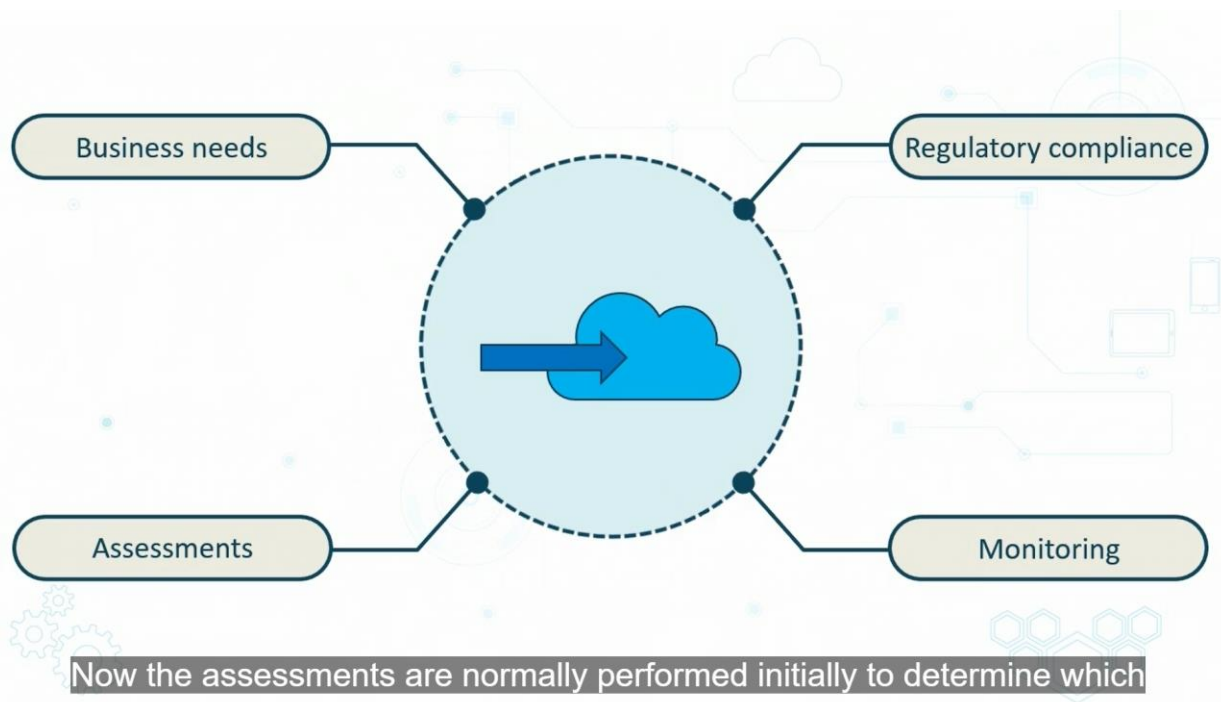
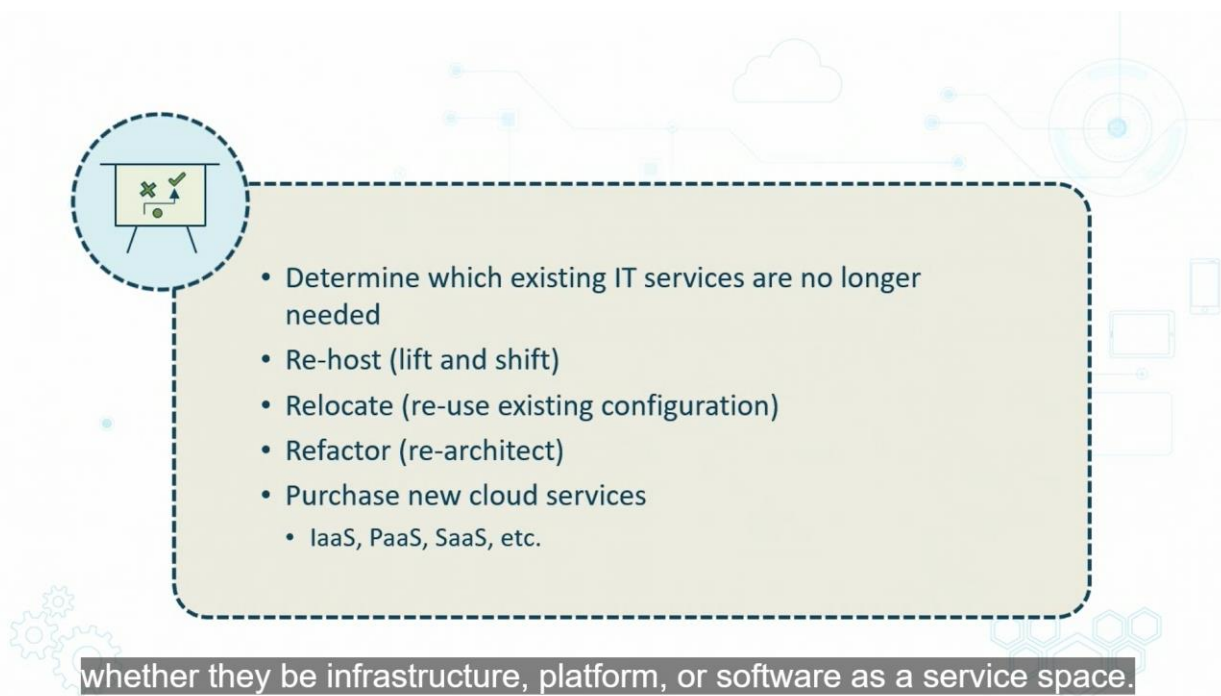


4.5 Migration Strategies

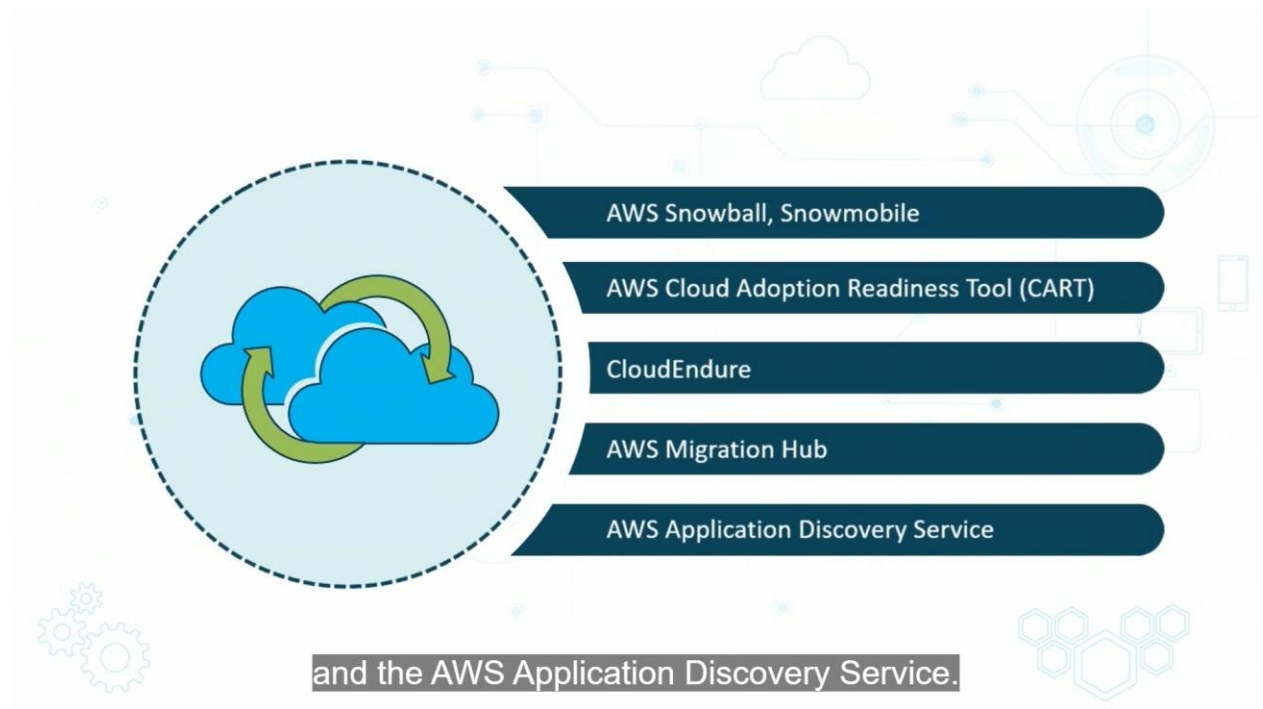
Cloud Migration Options



AWS Migration Strategies



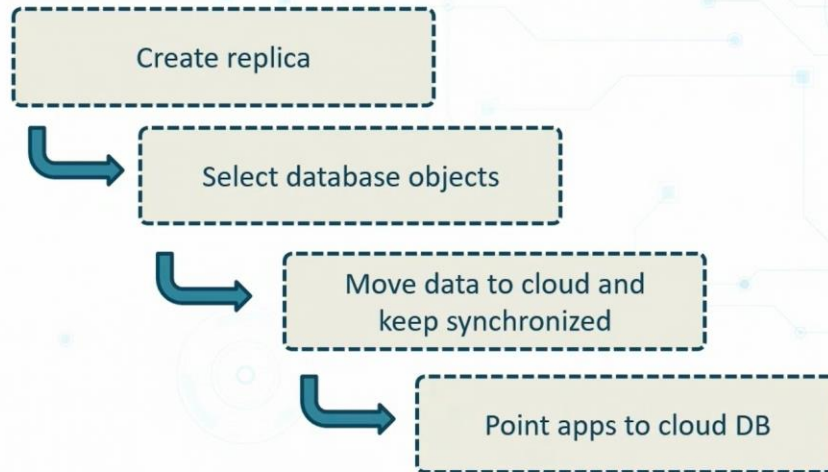
AWS Migration Tools



Database Migration

- A heterogeneous type of database migration means you're migrating it from one on-premises database platform to a different one in the cloud while retaining the data.
- Where a homogeneous type of migration means that if you were running MySQL on-premises, you can migrate that directly to MySQL in the cloud. It's the same database platform.

Amazon Database Migration Service (DMS)

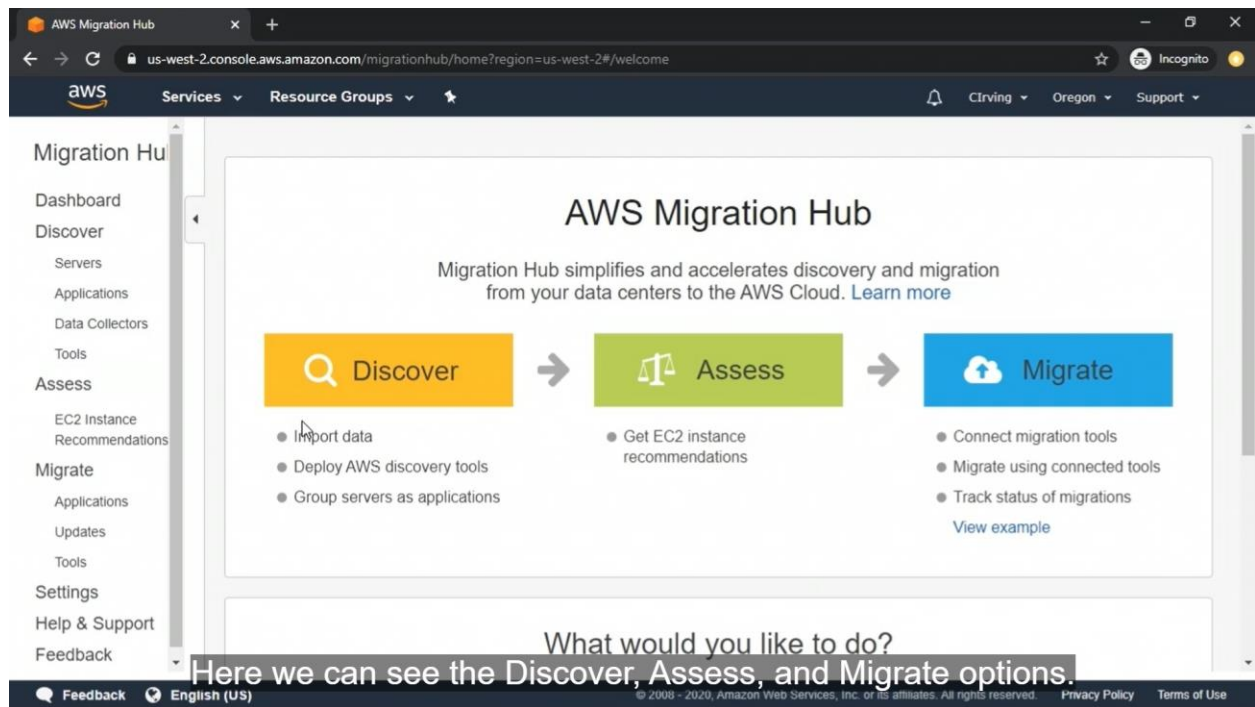


Of course, once that's done

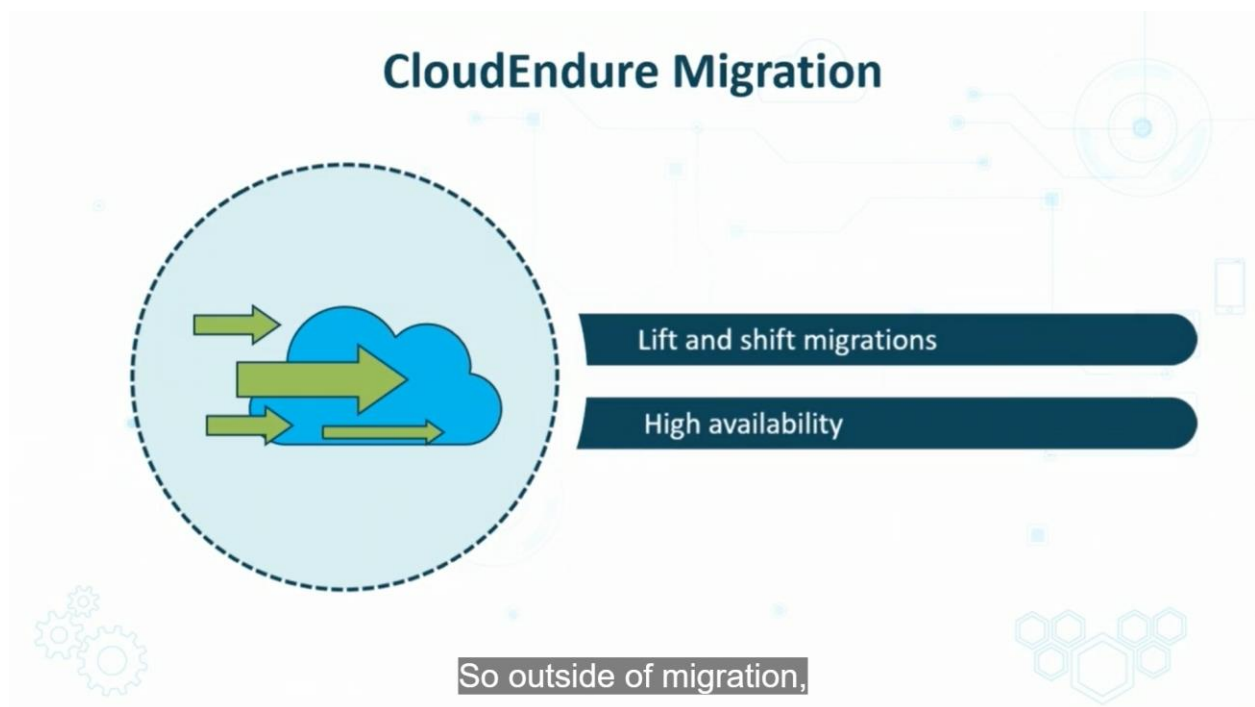
Cloud Migration Assessments

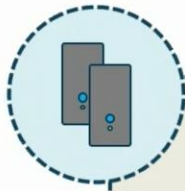
The screenshot shows the AWS Cloud Adoption Readiness Tool (CART) website. The browser address bar displays "cart.splashthat.com". The AWS logo is in the top left, with links for "Start Assessment" and "Additional Resources". The main heading is "AWS Cloud Adoption Readiness Tool (CART)" with the subtext "Assess your organization's cloud readiness in minutes." Below this is a large orange button labeled "Start Assessment". At the bottom, a section titled "Set your path for cloud adoption success" states: "CART helps organizations of all sizes develop a plan for cloud migration and adoption. Answer 16 questions about an organization's attitude and posture as related to cloud adoption. of: business, people, process, platform, operations, and security."

AWS Migration Hub



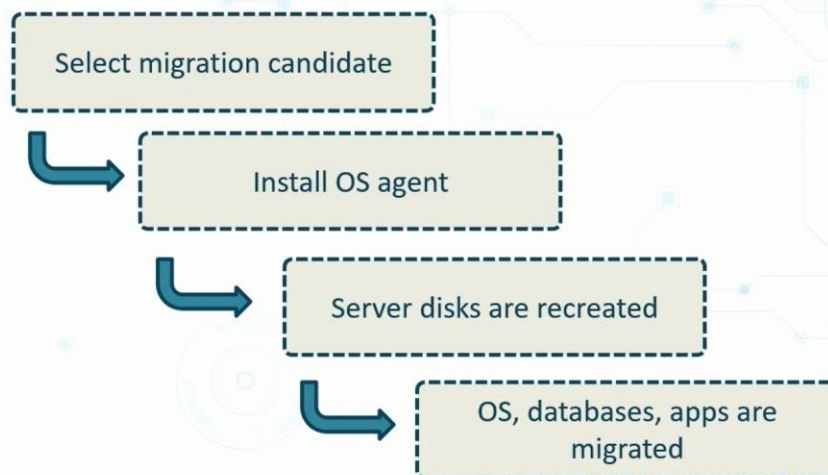
CloudEndure Migrations





- Windows and Linux on-prem hosts
 - Physical and virtual
- Target machine scripts
 - /boot/post_launch
 - C:\program files (x86)\cloudendure\post_launch
- Manage the migration using the CloudEndure console

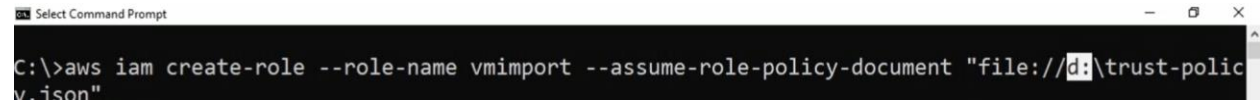
Server Migration Using CloudEndure



And then finally the OS, any databases, and

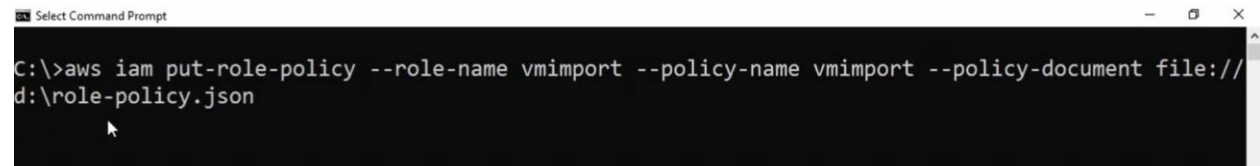
VM Image Prerequisites

Create role



```
C:\>aws iam create-role --role-name vmimport --assume-role-policy-document "file://d:\trust-policy.json"
```

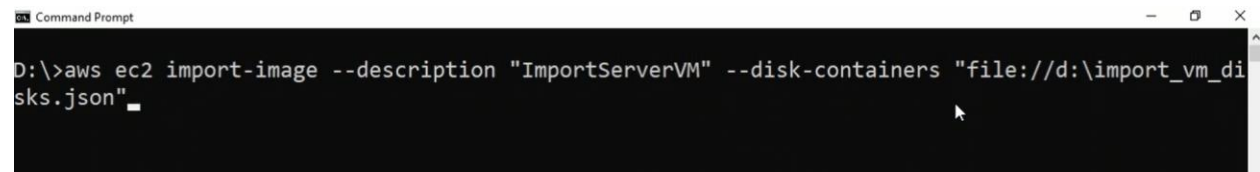
Put role policy



```
C:\>aws iam put-role-policy --role-name vmimport --policy-name vmimport --policy-document file://d:\role-policy.json
```

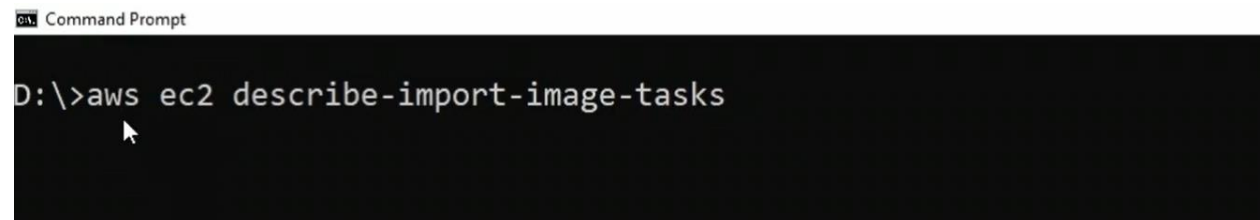
VM Image Import

Import image



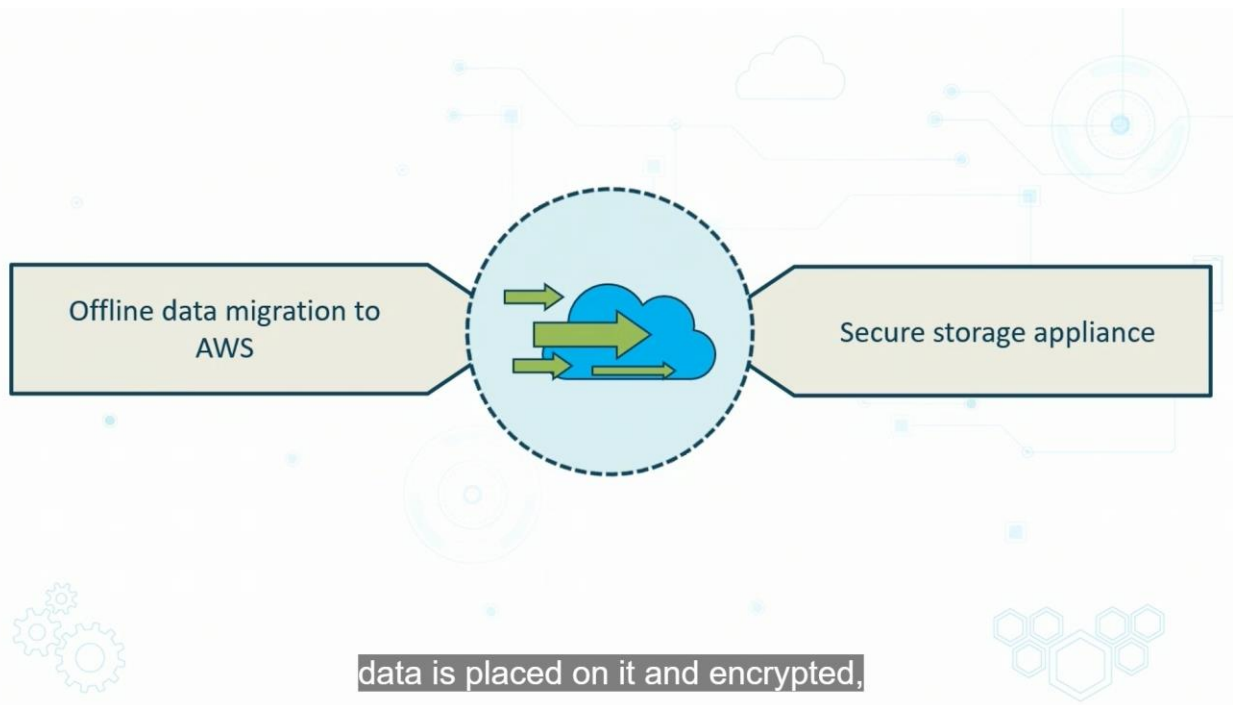
```
D:\>aws ec2 import-image --description "ImportServerVM" --disk-containers "file://d:\import_vm_disks.json"
```

Describe image



```
D:\>aws ec2 describe-import-image-tasks
```


AWS Snowball and Snowmobile



AWS Snowball

- Petabytes of on-premises data
- Create a "job" to have appliance sent to your location

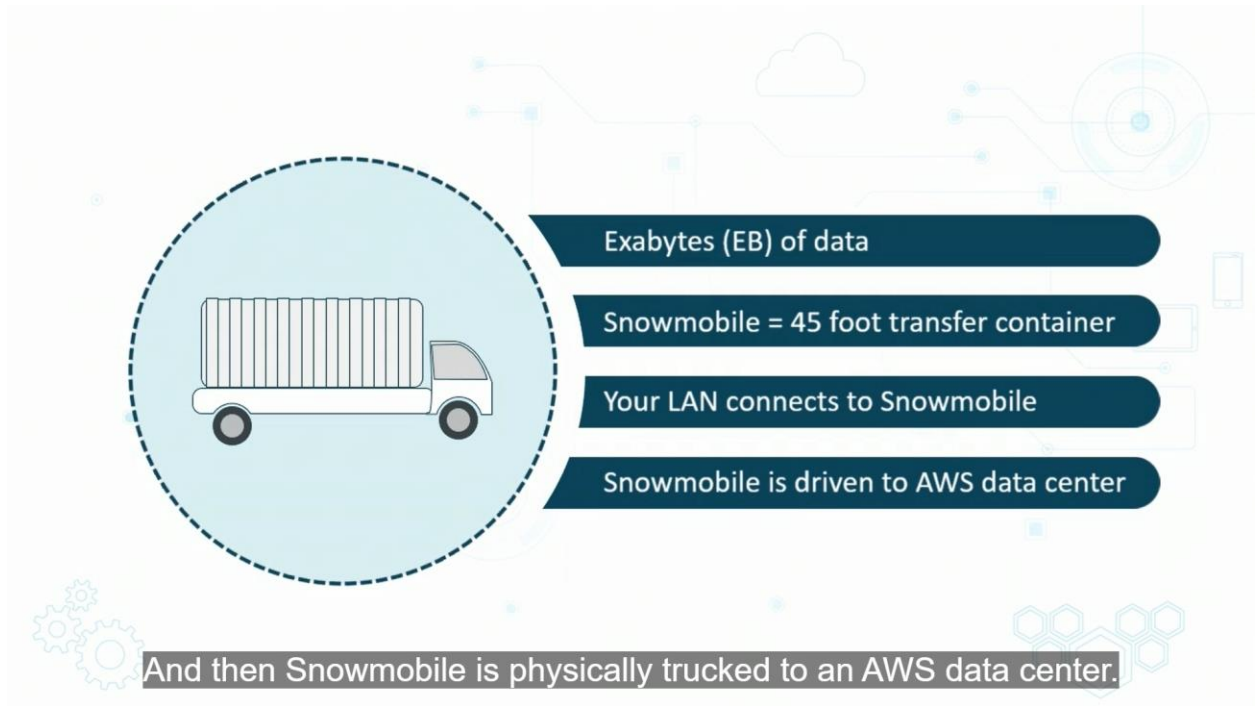


Send network appliances



So as an AWS technician, you would create a job – and you can do that

Snowmobile



AWS Snowball Jobs

1. You need to populate the AWS Migration Hub with on-premises server information. Which of the following could be used to accomplish this?
 - Download discovery agent
 - Import CSV file
2. An on-premises custom line of business application cannot be moved directly into the cloud without significant changes. What type of migration strategy should you use for this application?
 - Refactoring
3. What does the CART migration tool consist of?
 - Questionnaire

4. Which CLI command is used to import a VM disk image to S3?
 - `aws ec2 import-image`

5. Which statement regarding CloudEndure migrations is correct?
 - The on-premises server agent does not require a reboot

6. To which type of AWS resource does the `vmimport` role require permissions?
 - S3

7. When should AWS Snowball be used?
 - PB or TB of on-premises data that needs to be placed in AWS

8. Which statements regarding AWS Snowball jobs are correct?
 - KMS can be used to protect data at rest
 - A shipping address must be specified